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An exact solution to layered transversely isotropic poroelastic media under vertical rectangular moving loads

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A B S T R A C T

In this paper, we investigate the response of multilayered transversely isotropic poroelastic media under vertical rectangular moving loads by utilizing the extended precise integration method (PIM). Firstly, governing equations are derived based on Biot's formulation of porous media in the Cartesian coordinate system. Next, the Fourier expansion and integral transform techniques are applied to obtain the ordinary differential equations of a single layer. By introducing the continuity and boundary conditions of each layer, the solution in the transformed domain for multilayered media is further formulated. Then, the real solution is obtained by inverting the Fourier transform. Finally, the proposed solution is verified by comparing with existing solutions, and the impacts of transverse isotropy, anisotropic permeability, medium stratification and moving speed on the dynamic response are analyzed.

1. Introduction

Dynamic response of poroelastic media under moving loads is an essential problem in civil engineering, seismic engineering and transportation engineering, such as the vibration and deformation caused by cars, trains or subways. Therefore, research on this topic has been studied by many scholars. Owing to the difficulty of this problem, many investigations treated natural soils as elastic media for convenience. Sneddon (1951) first obtained the solution of the low-speed moving load acting on an elastic half-space. Then, Cole and Huth (1958) deduced the solutions of a concentrated moving load applied on a half-plane when the load speed is subsonic, transonic or supersonic. Nevertheless, Georgiadis and Barber (1993) found that the solution of Cole and Huth (1958) was not accurate in the transonic range due to an incorrect separation of their complex potentials into real and imaginary parts. Subsequently, Rahman (2001) utilized the Fourier transform to extend the solution given by Cole and Huth (1958) to the transonic case. Apart from the half-plane or half-space (Eason, 1965; Gakenheimer and Miklowitz, 1969), the layered model is more widely accepted by many scholars (de Barros and Luco, 1994; de Barros and Luco, 1995; Kim, 1998; Jin et al., 2013; Ai and Ren, 2016; Ai et al., 2017). With the consideration of the stratification, a series of methods, such as the Green's function method (Kim, 1998), the thin-layer method (Jin et al., 2013), the analytical layer-element method (Ai and Ren, 2016) and the extended precise integration method (PIM) (Ai et al., 2017), are utilized to investigate the dynamic response of layered elastic media excited by moving loads.

Actually, it is found that the existence of pore fluid in poroelastic media has a significant influence on the propagation of transverse and longitudinal waves, as well as the vibration and the deformation of the media (Cieszko and Kubik, 2020). It is more reasonable to treat the soils as poroelastic media in many practical engineering. Numerous researches are devoted to the field of the dynamic response of the poroelastic half-space (Biot, 1956a, 1956b; Jones, 1961; Mei and Foda, 1981; Ba et al., 2017). Based on the above achievements, the behaviors of poroelastic media due to moving loads have also drawn increasing attentions of the scholars (Burke and Kingsbury, 1984; Theodorakopoulos, 2003; Jin et al., 2004; Lu and Jeng, 2007; Lefeuve-Mesgouez and Mesgouez, 2008). For example, Theodorakopoulos (2003) investigated the plane strain problems of moving loads acting on a poroelastic soil medium. Furthermore, by introducing a moving coordinate system, Jin et al. (2004) deduced the solution of a poroelastic half-space induced by steady-state moving loads. Lu and Jeng (2007) presented an analytical solution by Dirac function for the moving point force applying on a poroelastic half-space. Moreover, Lefeuve-Mesgouez and Mesgouez (2008) analyzed the behaviors of viscoelastic saturated media caused by moving harmonic loads and gave the results when the load speed

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Nomenclature

$\sigma_x, \sigma_y, \sigma_z$	normal stress component in *x*, *y*, *z* directions
$\sigma_{xy}, \sigma_{yz}, \sigma_{xz}$	shear stress component in *x*-*y*, *y*-*z* and *x*-*z* planes
u_x, u_y, u_z	displacement component of solid skeleton in *x*, *y*, *z* directions
v_x, v_y, v_z	relative displacement component of pore fluid to solid skeleton in *x*, *y*, *z* directions
$c_{11}, c_{12}, c_{13}, c_{33}, c_{44}, c_{66}$	elastic constants of transversely isotropic solid skeleton
ρ_f, ρ_s	pore fluid density and solid skeleton density
g	gravity acceleration
ρ	density for saturated porous elastomer
φ	porosity
α_1, α_3	compressibility coefficients of solid skeleton in horizontal and vertical directions
$E_v, E_h, G_v, \mu_h, \mu_{vh}$	parameters of transversely isotropic solid skeleton
p	excess pore fluid pressure
$a_{\infty 1}, a_{\infty 3}$	tortuosities in horizontal and vertical directions
K_s, K_f	bulk moduli of solid skeleton and pore fluid
ς	material damping ratio
k_1, k_3	horizontal and vertical permeability coefficients
M	coefficients related to compressibility of solid skeleton and pore fluid
q_x, q_y, q_z	pore fluid unit flow in *x*, *y*, *z* directions
Q_f	pore fluid flow in *z* direction
F	magnitude of moving loads
v	velocity of moving loads
ξ	Fourier transform parameter
Superscript -	variables in Fourier transformed domain
$n = E_h/E_v$	ratio of horizontal Young's modulus to vertical Young's modulus
$\kappa = k_1/k_3$	ratio of horizontal permeability coefficient to vertical permeability coefficient

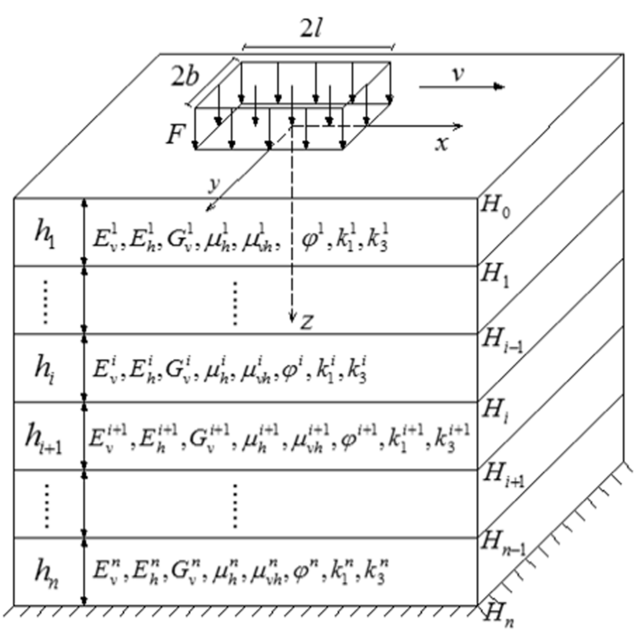

Fig. 1. A layered transversely isotropic poroelastic medium in Cartesian coordinates excited by a moving load with a constant speed.

exceeds the Rayleigh wave.

The research mentioned above is limited to a finite layer or a single-layered half-space. However, the multilayered model is more suitable for practical applications. In terms of the multilayered poroelastic media, Siddharthan et al. (1993a, 1993b) used Fourier series expansion to describe the moving loads, and extended the approach to layered poroelastic media under different types of moving load. Lefeuve-Mesgouez and Mesgouez (2012) proposed a semi-analytical solution for the layered porous viscoelastic media excited by harmonic moving loads. Moreover, Ba and Liang (2017) utilized the method of dynamic stiffness matrix and Fourier transform to acquire the steady-state analytical solution when moving point loads and pore pressure acts on layered transversely isotropic porous media.

From the literature mentioned above, we can find that few researchers have attempted to study the response of multilayered transversely isotropic poroelastic media under rectangular moving loads. Hallonborg (1996) pointed that the form of traffic loads are closely related to tyre shape, inflation pressure and soil parameters. In terms of the traffic load generated by small cars using low-pressure tires and vehicles moving on soft soils, the rectangular load is a commonly simplified form in the mathematical modelling. Hence, this paper focuses on the rectangular load.

Based on Biot's poroelastodynamic theory (Biot, 1956a, 1956b), we first establish the governing equations of the media, and then apply the Fourier series expansion to process the moving load. Accordingly, the ordinary differential matrix equation of single-layered media is obtained through the Fourier transform. With considering the boundary conditions and using the extended PIM of layered media, the solution in the transformed domain of the equation is gotten. By using the inversion of Fourier transform, the solution in the physical domain can be derived. Finally, the proposed solution is verified by several examples. In addition, several numerical examples are developed to discuss the influence of the factors such as transverse isotropy, anisotropic permeability, medium stratification and moving load speed on the dynamic response of the media. The moving load speeds in this paper are all in the subsonic range.

2. Ordinary differential equations for transversely isotropic poroelastic media

2.1. Basic equations

As show in Fig. 1, the poroelastic media consist of several layers with different properties, and a rectangular moving load F with an area of $2l \times 2b$ acts on the surface. A Cartesian coordinate system is established with its origin at the load centre position. The load moves with a constant speed v along the x-axis. The shear wave speed is defined as $\sqrt{G/\rho}$ (Singh, 2010), and the speed v is slower than it here.

According to Biot's poroelastodynamic theory (Biot, 1956a), the equilibrium equations of porous media in Cartesian coordinates are

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = \rho \frac{\partial^2 u_x}{\partial t^2} + \rho_f \frac{\partial^2 v_x}{\partial t^2} \tag{1}$$

$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = \rho \frac{\partial^2 u_y}{\partial t^2} + \rho_f \frac{\partial^2 v_y}{\partial t^2} \tag{2}$$

$$\frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_z}{\partial z} = \rho \frac{\partial^2 u_z}{\partial t^2} + \rho_f \frac{\partial^2 v_z}{\partial t^2} \tag{3}$$

where σ_x, σ_y, σ_z are the normal stresses in the x,y, z directions, respectively; σ_{xy}, σ_{yz}, σ_{xz} denote the shear stresses in the x-y, y-z and x-z planes, respectively; u_x, u_y, u_z denote the displacement components of solid skeleton in x, y, z directions, respectively; v_x, v_y, v_z are the relative displacements of fluid and solid skeleton in x, y, z directions, respectively; ρ, ρ_f are the densities of poroelastic media and pore fluid which have the relationship as $\rho = \varphi\rho_f + (1-\varphi)\rho_s$, where ρ_s is the solid skeleton density and φ is the medium porosity.

Considering the principle of effective stress, we have the constitutive equations of the poroelastic medium:

$$\sigma_x = c_{11}\frac{\partial u_x}{\partial x} + c_{12}\frac{\partial u_y}{\partial y} + c_{13}\frac{\partial u_z}{\partial z} - \alpha_1 p \tag{4}$$

$$\sigma_y = c_{12}\frac{\partial u_x}{\partial x} + c_{11}\frac{\partial u_y}{\partial y} + c_{13}\frac{\partial u_z}{\partial z} - \alpha_1 p \tag{5}$$

$$\sigma_z = c_{13}\frac{\partial u_x}{\partial x} + c_{13}\frac{\partial u_y}{\partial y} + c_{33}\frac{\partial u_z}{\partial z} - \alpha_3 p \tag{6}$$

$$\sigma_{xz} = c_{44}\left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x}\right) \tag{7}$$

$$\sigma_{yz} = c_{44}\left(\frac{\partial u_y}{\partial z} + \frac{\partial u_z}{\partial y}\right) \tag{8}$$

$$\sigma_{xy} = c_{66}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) \tag{9}$$

where $c_{11} = \lambda n(1-n\mu_{vh}^2)$, $c_{12} = \lambda n(\mu_h + n\mu_{vh}^2)$, $c_{13} = \lambda n\mu_{vh}(1+\mu_h)$, $c_{33} = \lambda(1-\mu_h^2)$, $c_{44} = G_v$, $c_{66} = (c_{11}-c_{12})/2$, $\lambda = E_v/[(1+\mu_h)(1-\mu_h-2n\mu_{vh}^2)]$, $n = E_h/E_v$; p is the excess pore fluid pressure. Both pore fluid and solid skeleton are compressible. α_1 and α_3 are the solid skeleton compressibility coefficient in horizontal and vertical directions and can be expressed as

$$\alpha_1 = 1 - \frac{2c_{11} - 2c_{66} + c_{13}}{3K_s} \tag{10}$$

$$\alpha_3 = 1 - \frac{2c_{13} + c_{33}}{3K_s} \tag{11}$$

in which K_s is the bulk modulus of solid skeleton.

The singularity of the integrand function sometimes causes the integral not to converge. Some scholars used the residual theory of integration to solve this problem (Khojasteh et al., 2008, 2011). In this paper, in order to avoid the influence of the singularity, a small damping term is added (Ai et al., 2014). Material damping can be introduced by letting $c_{11}^* = c_{11}(1+2\mathrm{i}\varsigma)$, $c_{12}^* = c_{12}(1+2\mathrm{i}\varsigma)$, $c_{13}^* = c_{13}(1+2\mathrm{i}\varsigma)$, $c_{33}^* = c_{33}(1+2\mathrm{i}\varsigma)$ and $c_{44}^* = c_{44}(1+2\mathrm{i}\varsigma)$, here $\varsigma = 0.01$ suggested by Ba and Liang (2017).

According to Biot's poroelastodynamic theory, we have

$$-\frac{\partial p}{\partial x} = \rho_f\frac{\partial^2 u_x}{\partial t^2} + \frac{\rho_f g}{k_1}\frac{\partial v_x}{\partial t} + \frac{a_{\infty 1}\rho_f}{\varphi}\frac{\partial^2 v_x}{\partial t^2} \tag{12}$$

$$-\frac{\partial p}{\partial y} = \rho_f\frac{\partial^2 u_y}{\partial t^2} + \frac{\rho_f g}{k_1}\frac{\partial v_y}{\partial t} + \frac{a_{\infty 1}\rho_f}{\varphi}\frac{\partial^2 v_y}{\partial t^2} \tag{13}$$

$$-\frac{\partial p}{\partial z} = \rho_f\frac{\partial^2 u_z}{\partial t^2} + \frac{\rho_f g}{k_3}\frac{\partial v_z}{\partial t} + \frac{a_{\infty 3}\rho_f}{\varphi}\frac{\partial^2 v_z}{\partial t^2} \tag{14}$$

where k_1 and k_3 are the horizontal and vertical permeability coefficients, respectively; $a_{\infty 1}$ denotes horizontal dynamic tortuosity, and $a_{\infty 3}$ denotes vertical dynamic tortuosity.

For dynamic problems, the expression of excess pore fluid pressure is (Chen et al., 2018)

$$-p = \alpha_1 M\left(\frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y}\right) + \alpha_3 M\frac{\partial u_z}{\partial z} + M\left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}\right) \tag{15}$$

where

$$\frac{1}{M} = \frac{1-\varphi}{K_s} + \frac{\varphi}{K_f} - \frac{4c_{11} - 4c_{66} + 4c_{13} + c_{33}}{9K_s^2} \tag{16}$$

in which M is the coefficient related to the compressibility of pore fluid and solid skeleton, and K_f is the bulk modulus of fluid.

Based on the Darcy's law, the total amount of seepage in 0-t time period along z direction can be obtained by

$$Q_f = -\int_0^t \frac{k_3}{\rho_f g}\frac{\partial p}{\partial z}\mathrm{d}t \tag{17}$$

2.2. Load forms and governing equations in the transformed domain

The moving load is assumed to be expanded in Fourier series. A load distribution of any shape can be described by such an expansion. According to the references (Theodorakopoulos, 2003), the uniform moving load can be written as

$$F(x,y) = \begin{cases} F & |x|\leqslant l, |y|\leqslant b \\ 0 & l < |x| < L \end{cases} \tag{18}$$

By utilizing the Fourier series expansion, the uniform moving load can be expressed as

$$F(x-vt,y) = Re\sum_{-\infty}^{+\infty} F_n \mathrm{e}^{\mathrm{i}\lambda_n(x-vt)} \quad |y|\leqslant b \tag{19}$$

$$F_n = \begin{cases} \dfrac{Fl}{L} & n = 0 \\ \dfrac{2F}{n\pi}\sin\dfrac{n\pi l}{L} & n > 0 \end{cases} \quad |y|\leqslant b \tag{20}$$

where F denotes the moving load, $\lambda_n = n\pi/L$ and $\mathrm{i} = \sqrt{-1}$, and Re means taking the real part of the complex number.

Similarly, when it refers to the concentrated moving load, it can be expressed by using the Fourier series expansion as

$$F^c(x-vt,y) = Re\sum_{-\infty}^{+\infty} F_n^c(y)\mathrm{e}^{\mathrm{i}\lambda_n(x-vt)} \tag{21}$$

$$F_n^c(y) = \frac{F^c\delta(y)}{2L} \tag{22}$$

where F^c denotes the moving load and $\delta(\cdot)$ is the Dirac function.

When the external load is expressed by the Eqs. (19) and (20) or Eqs. (21) and (22), the response function of the poroelastic medium under moving loads can be expressed as

$$\phi(x-vt,y,z,t) = Re\sum_{-\infty}^{+\infty}\phi_n(y,z)\mathrm{e}^{\mathrm{i}\lambda_n(x-vt)} \tag{23}$$

where $\phi_n(y,z)$ is just related to y and z while x and t only appear in the index term. Since the exponential function $\mathrm{e}^{\mathrm{i}\lambda_n(x-vt)}$ appears in each term, it is omitted in the following derivation.

The partial differentiation of any term of the series of response function to x, y, z can be written as

$$\begin{aligned} &\frac{\partial\phi}{\partial x} = \mathrm{i}\lambda_n\phi_n(y,z), \quad \frac{\partial\phi}{\partial t} = -\mathrm{i}\lambda_n v\phi_n(y,z), \\ &\frac{\partial\phi}{\partial y} = \frac{\partial\phi_n(y,z)}{\partial y}, \quad \frac{\partial\phi}{\partial z} = \frac{\partial\phi_n(y,z)}{\partial z} \end{aligned} \tag{24}$$

$$\frac{\partial^2\phi}{\partial x^2}=-\lambda_n^2\phi_n(y,z),\frac{\partial^2\phi}{\partial t^2}=-\lambda_n^2v^2\phi_n(y,z),$$
$$\frac{\partial^2\phi}{\partial x\partial t}=\lambda_n^2 v\phi_n(y,z) \tag{25}$$

According to Eqs. (24) and (25), Eqs. (7), (8) and (6) can be expressed as

$$\frac{\partial u_x}{\partial z}=\frac{1}{c_{44}}\sigma_{xz}-\mathrm{i}\lambda_n u_z \tag{26}$$

$$\frac{\partial u_y}{\partial z}=\frac{1}{c_{44}}\sigma_{yz}-\frac{\partial u_z}{\partial y} \tag{27}$$

$$\frac{\partial u_z}{\partial z}=\frac{1}{c_{33}}\sigma_z-\frac{\mathrm{i}\lambda_n c_{13}}{c_{33}}u_x-\frac{c_{13}}{c_{33}}\frac{\partial u_y}{\partial y}+\frac{\alpha_3}{c_{33}}p \tag{28}$$

By combining Eqs. (1), (4), (9) and (28), the following equation can be obtained as

$$\frac{\partial\sigma_{xz}}{\partial z}=\left(-\rho v^2\lambda_n^2+\lambda_n^2c_{11}-\frac{\lambda_n^2c_{13}^2}{c_{33}}\right)u_x-c_{66}\frac{\partial^2u_x}{\partial y^2}-\mathrm{i}\lambda_n\left(c_{12}+c_{66}-\frac{c_{13}^2}{c_{33}}\right)\frac{\partial u_y}{\partial y}$$
$$-\frac{\mathrm{i}\lambda_n c_{13}}{c_{33}}\sigma_z+\mathrm{i}\lambda_n\left(\alpha_1-\frac{\alpha_3c_{13}}{c_{33}}\right)p-\rho_f v^2\lambda_n^2 v_x \tag{29}$$

Similarly, we can obtain the following expression by combining Eqs. (2), (5), (9) and (28):

$$\frac{\partial\sigma_{yz}}{\partial z}=-\mathrm{i}\lambda_n\left(c_{12}+c_{66}-\frac{c_{13}^2}{c_{33}}\right)\frac{\partial u_x}{\partial y}+\left(-\rho v^2\lambda_n^2+\lambda_n^2c_{66}\right)u_y-\rho_f v^2\lambda_n^2 v_y$$
$$-\left(c_{11}-\frac{c_{13}^2}{c_{33}}\right)\frac{\partial^2u_y}{\partial y^2}-\frac{c_{13}}{c_{33}}\frac{\partial\sigma_z}{\partial y}+\left(\alpha_1-\frac{\alpha_3c_{13}}{c_{33}}\right)\frac{\partial p}{\partial y} \tag{30}$$

According to Eq. (3), we have

$$\frac{\partial\sigma_z}{\partial z}=-\rho v^2\lambda_n^2u_z-\rho_f v^2\lambda_n^2v_z-\mathrm{i}\lambda_n\sigma_{xz}-\frac{\partial\sigma_{yz}}{\partial y} \tag{31}$$

Based on Eqs. (12)–(14), the following equations can be derived as

$$v_x=c_1\left(\mathrm{i}\lambda_n p-\rho_f v^2\lambda_n^2u_x\right) \tag{32}$$

$$v_y=c_1\left(\frac{\partial p}{\partial y}-\rho_f v^2\lambda_n^2u_y\right) \tag{33}$$

$$v_z=c_3\left(\frac{\partial p}{\partial z}-\rho_f v^2\lambda_n^2u_z\right) \tag{34}$$

where $c_1=k_1\varphi/\left(\mathrm{i}\lambda_n v\rho_f g\varphi+\rho_f v^2\lambda_n^2a_{\infty1}k_1\right)$, and $c_3=k_3\varphi/\left(\mathrm{i}\lambda_n v\rho_f g\varphi+\rho_f v^2\lambda_n^2a_{\infty3}k_3\right)$.

Combining Eqs. (32)–(34), (14) and (15), we have

$$-\frac{\partial^2p}{\partial z^2}=\frac{\mathrm{i}\lambda_n}{c_3}\left(b_1-\frac{b_3c_{13}}{c_{33}}\right)u_x+\frac{1}{c_3}\left(b_1-\frac{b_3c_{13}}{c_{33}}\right)\frac{\partial u_y}{\partial y}+\frac{b_3}{c_3c_{33}}\sigma_z$$
$$+\frac{1}{c_3}\left(\frac{\alpha_3b_3}{c_{33}}+\frac{1}{M}-\lambda_n^2c_1\right)p+\frac{c_1}{c_3}\frac{\partial^2p}{\partial y^2} \tag{35}$$

where $b_1=\alpha_1-c_1\rho_f v^2\lambda_n^2$, and $b_3=\alpha_3-c_3\rho_f v^2\lambda_n^2$.

According to Eq. (17) and Eq. (24), the following equation can be obtained as

$$\frac{\partial p}{\partial z}=\frac{\mathrm{i}\lambda_n v\rho_f g}{k_3}Q_f \tag{36}$$

Combining Eqs. (35) and (36), we have

$$\frac{\partial Q_f}{\partial z}=-\frac{k_3}{\mathrm{i}\lambda_n v\rho_f gc_3}\left[\mathrm{i}\lambda_n\left(b_1-\frac{b_3c_{13}}{c_{33}}\right)u_x+\left(b_1-\frac{b_3c_{13}}{c_{33}}\right)\frac{\partial u_y}{\partial y}+\frac{b_3}{c_{33}}\sigma_z\right.$$
$$\left.+\left(\frac{\alpha_3b_3}{c_{33}}+\frac{1}{M}-\lambda_n^2c_1\right)p+c_1\frac{\partial^2p}{\partial y^2}\right] \tag{37}$$

Substituting Eqs. (32)–(34) into Eqs. (29)–(31), we have

$$\frac{\partial\sigma_{xz}}{\partial z}=\lambda_n^2\left(-\rho v^2+c_{11}-\frac{c_{13}^2}{c_{33}}+\rho_f^2v^4\lambda_n^2c_1\right)u_x-c_{66}\frac{\partial^2u_x}{\partial y^2}$$
$$-\mathrm{i}\lambda_n\left(c_{12}+c_{66}-\frac{c_{13}^2}{c_{33}}\right)\frac{\partial u_y}{\partial y}-\frac{\mathrm{i}\lambda_nc_{13}}{c_{33}}\sigma_z+\mathrm{i}\lambda_n\left(\alpha_1-\frac{\alpha_3c_{13}}{c_{33}}-\rho_f v^2\lambda_n^2c_1\right)p \tag{38}$$

$$\frac{\partial\sigma_{yz}}{\partial z}=-\mathrm{i}\lambda_n\left(c_{12}+c_{66}-\frac{c_{13}^2}{c_{33}}\right)\frac{\partial u_x}{\partial y}+\lambda_n^2\left(-\rho v^2+c_{66}+\rho_f^2v^4\lambda_n^2c_1\right)u_y$$
$$-\left(c_{11}-\frac{c_{13}^2}{c_{33}}\right)\frac{\partial^2u_y}{\partial y^2}-\frac{c_{13}}{c_{33}}\frac{\partial\sigma_z}{\partial y}+\left(\alpha_1-\frac{\alpha_3c_{13}}{c_{33}}-\rho_f v^2\lambda_n^2c_1\right)\frac{\partial p}{\partial y} \tag{39}$$

$$\frac{\partial\sigma_z}{\partial z}=\lambda_n^2\left(-\rho v^2+\rho_f^2v^4\lambda_n^2c_3\right)u_z-\mathrm{i}\lambda_n\sigma_{xz}-\frac{\partial\sigma_{yz}}{\partial y}-\frac{\mathrm{i}\rho_f^2v^3\lambda_n^3gc_3}{k_3}Q_f \tag{40}$$

The integral transformation method is introduced to eliminate the dependency of variables y in the partial differential equations. The definitions of Fourier transform and its inverse transform are as follows (Sneddon, 1951; Chen et al., 2018):

$$\bar f(x,\xi_y,z,t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}f(x,y,z,t)\mathrm{e}^{-\mathrm{i}\xi_y y}\mathrm{d}y \tag{41}$$

$$f(x,y,z,t)=\int_{-\infty}^{+\infty}\bar f(x,\xi_y,z,t)\mathrm{e}^{\mathrm{i}\xi_y y}\mathrm{d}\xi_y \tag{42}$$

where $\bar f(x,\xi_y,z,t)$ represents the function of $f(x,y,z,t)$ after Fourier transform, and ξ_y is the Fourier transform parameter regarding y.

By using Fourier transforms, Eqs. (26)–(28), (36), (37) and (38)–(40) can further be expressed as

$$\frac{\partial\bar u_x}{\partial z}=\frac{1}{c_{44}}\bar\sigma_{xz}-\mathrm{i}\lambda_n\bar u_z \tag{43}$$

$$\frac{\partial\bar u_y}{\partial z}=\frac{1}{c_{44}}\bar\sigma_{yz}-\mathrm{i}\xi_y\bar u_z \tag{44}$$

$$\frac{\partial\bar u_z}{\partial z}=\frac{1}{c_{33}}\bar\sigma_z-\frac{\mathrm{i}\lambda_nc_{13}}{c_{33}}\bar u_x-\frac{\mathrm{i}\xi_yc_{13}}{c_{33}}\bar u_y+\frac{\alpha_3}{c_{33}}\bar p \tag{45}$$

$$\frac{\partial\bar\sigma_{xz}}{\partial z}=-\frac{\mathrm{i}\lambda_nc_{13}}{c_{33}}\bar\sigma_z+\left(-\rho v^2\lambda_n^2+c_{11}\lambda_n^2-\frac{\lambda_n^2c_{13}^2}{c_{33}}+\rho_f^2v^4\lambda_n^4c_1+c_{66}\xi_y^2\right)\bar u_x$$
$$+\xi_y\lambda_n\left(c_{12}+c_{66}-\frac{c_{13}^2}{c_{33}}\right)\bar u_y+\mathrm{i}\lambda_n\left(\alpha_1-\frac{\alpha_3c_{13}}{c_{33}}-\rho_f v^2\lambda_n^2c_1\right)\bar p \tag{46}$$

$$\frac{\partial\bar\sigma_{yz}}{\partial z}=-\frac{\mathrm{i}\xi_yc_{13}}{c_{33}}\bar\sigma_z+\left(-\rho v^2\lambda_n^2+c_{66}\lambda_n^2+\rho_f^2v^4\lambda_n^4c_1+\xi_y^2c_{11}-\frac{\xi_y^2c_{13}^2}{c_{33}}\right)\bar u_y$$
$$+\xi_y\lambda_n\left(c_{12}+c_{66}-\frac{c_{13}^2}{c_{33}}\right)\bar u_x+\mathrm{i}\xi_y\left(\alpha_1-\frac{\alpha_3c_{13}}{c_{33}}-\rho_f v^2\lambda_n^2c_1\right)\bar p \tag{47}$$

$$\frac{\partial\bar\sigma_z}{\partial z}=-\mathrm{i}\lambda_n\bar\sigma_{xz}-\mathrm{i}\xi_y\bar\sigma_{yz}+\lambda_n^2\left(-\rho v^2+\rho_f^2v^4\lambda_n^2c_3\right)\bar u_z-\frac{\mathrm{i}\rho_f^2v^3\lambda_n^3gc_3}{k_3}\bar Q_f \tag{48}$$

$$\frac{\partial\bar p}{\partial z}=\frac{\mathrm{i}\lambda_n v\rho_f g}{k_3}\bar Q_f \tag{49}$$

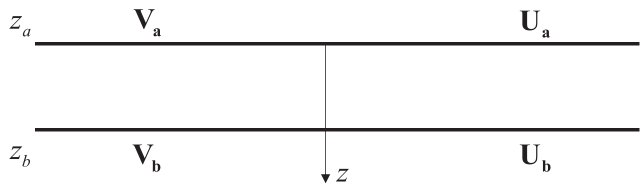

Fig. 2. State vectors of a single layer.

$$\frac{\partial \overline{Q}_f}{\partial z} = -\frac{k_3}{\mathrm{i}\lambda_n v \rho_f g c_3}\left[\mathrm{i}\lambda_n\left(b_1 - \frac{b_3 c_{13}}{c_{33}}\right)\overline{u}_x + \mathrm{i}\xi_y\left(b_1 - \frac{b_3 c_{13}}{c_{33}}\right)\overline{u}_y + \frac{b_3}{c_{33}}\overline{\sigma}_z + \left(\frac{\alpha_3 b_3}{c_{33}} + \frac{1}{M} - \lambda_n^2 c_1 - \xi_y^2 c_1\right)\overline{p}\right] \tag{50}$$

Eqs. (43)–(50) are also written as the following matrix form

$$\frac{\mathrm{d}}{\mathrm{d}z}\mathbf{W}(z) = \mathbf{H}\cdot\mathbf{W}(z) \tag{51}$$

where $\mathbf{W}(z) = [\mathbf{V} \quad \mathbf{U}]^T$, $\mathbf{V} = \begin{bmatrix}\overline{\sigma}_{xz} & \overline{\sigma}_{yz} & \overline{\sigma}_z & \overline{p}\end{bmatrix}^T$ is the vector of generalized stresses, and $\mathbf{U} = \begin{bmatrix}\overline{u}_x & \overline{u}_y & \overline{u}_z & \overline{Q}_f\end{bmatrix}^T$ is the vector of generalized displacements. $\mathbf{H} = \begin{bmatrix}\mathbf{A} & \mathbf{D}\\ \mathbf{B} & \mathbf{C}\end{bmatrix}$, among them, each element of the coefficient matrices $\mathbf{A}_{4\times4}$, $\mathbf{B}_{4\times4}$, $\mathbf{C}_{4\times4}$, $\mathbf{D}_{4\times4}$ are $\mathrm{A}_{13} = -\mathrm{i}\lambda_n c_{13}/c_{33}$, $\mathrm{A}_{14} = \mathrm{i}\lambda_n(\alpha_1 - \alpha_3 c_{13}/c_{33} - \rho_f v^2\lambda_n^2 c_1)$, $\mathrm{A}_{23} = -\mathrm{i}\xi_y c_{13}/c_{33}$, $\mathrm{A}_{24} = \mathrm{i}\xi_y(\alpha_1 - \alpha_3 c_{13}/c_{33} - \rho_f v^2\lambda_n^2 c_1)$, $\mathrm{A}_{31} = -\mathrm{i}\lambda_n$, $\mathrm{A}_{32} = -\mathrm{i}\xi_y$, $\mathrm{B}_{11} = 1/c_{44}$, $\mathrm{B}_{22} = 1/c_{44}$, $\mathrm{B}_{33} = 1/c_{33}$, $\mathrm{B}_{34} = \alpha_3/c_{33}$, $\mathrm{B}_{43} = -b_3k_3/(\mathrm{i}\lambda_n v\rho_f g c_{33} c_3)$, $\mathrm{B}_{44} = -k_3\left(\alpha_3 b_3/c_{33} + 1/M - \lambda_n^2 c_1 - \xi_y^2 c_1\right)\big/(\mathrm{i}\lambda_n v\rho_f g c_3)$, $\mathrm{C}_{13} = -\mathrm{i}\lambda_n$, $\mathrm{C}_{23} = -\mathrm{i}\xi_y$, $\mathrm{C}_{31} = -\mathrm{i}\lambda_n c_{13}/c_{33}$, $\mathrm{C}_{32} = -\mathrm{i}\xi_y c_{13}/c_{33}$, $\mathrm{C}_{41} = -\mathrm{i}\lambda_n k_3(b_1 - b_3 c_{13}/c_{33})/(\mathrm{i}\lambda_n v\rho_f g c_3)$, $\mathrm{C}_{42} = -\mathrm{i}\xi_y k_3(b_1 - b_3 c_{13}/c_{33})/(\mathrm{i}\lambda_n v\rho_f g c_3)$, $\mathrm{D}_{11} = \lambda_n^2(c_{11} - c_{13}^2/c_{33}) - \rho v^2\lambda_n^2 + \rho_f^2 v^4\lambda_n^4 c_1 + \xi_y^2 c_{66}$, $\mathrm{D}_{12} = \xi_y\lambda_n(c_{12} + c_{66} - c_{13}^2/c_{33})$, $\mathrm{D}_{21} = \xi_y\lambda_n(c_{12} + c_{66} - c_{13}^2/c_{33})$, $\mathrm{D}_{22} = \lambda_n^2 c_{66} - \rho v^2\lambda_n^2 + \rho_f^2 v^4\lambda_n^4 c_1 + \xi_y^2(c_{11} - c_{13}^2/c_{33})$, $\mathrm{D}_{33} = \rho_f^2 v^4\lambda_n^4 c_3 - \rho v^2\lambda_n^2$, $\mathrm{D}_{34} = -\mathrm{i}\rho_f^2 g v^3\lambda_n^3 c_3/k_3$, $\mathrm{D}_{44} = \mathrm{i}\lambda_n v\rho_f g/k_3$, respectively. The other elements of the coefficient matrices are 0.

When the boundary conditions are determined, the extended PIM can be used to solve Eq. (51) as shown in Cheng and Ai (2016) and Wang and Ai (2018).

3. Solutions for layered transversely isotropic poroelastic media

Fig. 2 shows the state vectors of a single layer. In fact, natural media usually consist of several layers with different properties, while Eq. (51) describes the generalized stresses and displacements on the top and bottom of any layer. If we regard Eq. (51) as a section ordinary differential matrix equation, the extended PIM can be further used to attain the response of multilayered transversely isotropic poroelastic media by combining with the continuity conditions between each layer.

The interfaces at the depths $H_i(i = 1, \ldots, n)$ are assumed to be perfectly bonded. The top of the medium is supposed to be free and permeable, so the generalized stress vector of the upper side is **0**. Similarly, the bottom of the medium is considered to be constant and impermeable, so the generalized displacement vector of the lower side is **0**. Then the boundary conditions in the transformed domain are

$$\left(\overline{\sigma}_{xz}\right)_n(\xi_y, 0) = 0,\ \left(\overline{\sigma}_{yz}\right)_n(\xi_y, 0) = 0,\ \left(\overline{\sigma}_z\right)_n(\xi_y, 0) = 0,\ (\overline{p})_n(\xi_y, 0) = 0 \tag{52}$$

$$\left(\overline{u}_x\right)_n(\xi_y, H_n) = 0,\ \left(\overline{u}_y\right)_n(\xi_y, H_n) = 0,\ \left(\overline{u}_z\right)_n(\xi_y, H_n) = 0,\ \left(\overline{Q}_f\right)_n(\xi_y, H_n) = 0 \tag{53}$$

As expressed in Eqs. (19)–(22), the analysis in this paper is based on the Fourier series expansion of the load. According to the discrete theory proposed by Siddharthan et al. (1993a), the load is dispersed into $2N+1$ identical spatial points in the $2L$ wavelength range. From 0 to $2N$, it should be ensured that the interval Δ_x is smaller than the load length $2l$ or the wavelength $2L$. In this paper, we assume that $\Delta_x = 0.1m$ and $2N = 4096$, then $2L = 409.6$m. Then the Eq. (19) and (21) can be further expressed as

$$F(x - vt, y) = Re\sum_{n=0}^{2048}\overline{F}_n \mathrm{e}^{\mathrm{i}\lambda_n(x - vt)} \tag{54}$$

where $\lambda_n = n\pi/L$, $|y| \leqslant b$.

When $n = 0$, $\lambda_n = 10^{-7}$, Eq. (23) can be further expressed as

$$\phi(x - vt, y, z, t) = Re\left[\phi_0(y, z) + \sum_{n=1}^{2048}\phi_n(y, z)\mathrm{e}^{\mathrm{i}\lambda_n(x - vt)}\right] \tag{55}$$

The uniform moving load in Fourier transformed domain can be expressed as

$$\overline{F}_n = \begin{cases}\dfrac{F\sin(\xi_y b)l}{\pi\xi_y L} & n = 0\\[2ex] \dfrac{2F\sin(\xi_y b)}{n\pi^2\xi_y}\sin\dfrac{n\pi l}{L} & n > 0\end{cases} \tag{56}$$

Similarly, the concentrated moving load in Fourier transformed domain can be written as

$$\overline{F}_n^c = \frac{F}{2\pi L},\quad n \geqslant 0 \tag{57}$$

Therefore, when a vertical moving load acts on the surface of a layer, the expression of the generalized load vector can be expressed as

$$\mathbf{F} = \begin{bmatrix}0 & 0 & \overline{F}_n & 0\end{bmatrix}^{\mathrm{T}} \tag{58}$$

or

$$\mathbf{F} = \begin{bmatrix}0 & 0 & \overline{F}_n^c & 0\end{bmatrix}^{\mathrm{T}} \tag{59}$$

The interfaces between adjacent layers are continuous, so the continuity condition between each layer can be written as

$$\mathbf{V}_\mathbf{i}(\xi_y, H_i^+) = \mathbf{V}_\mathbf{i}(\xi_y, H_i^-) + \mathbf{F}(\xi_y, H_i) \tag{60}$$

$$\mathbf{U}_\mathbf{i}(\xi_y, H_i^+) = \mathbf{U}_\mathbf{i}(\xi_y, H_i^-) \tag{61}$$

where $\mathbf{V}_\mathbf{i}(\xi_y, H_i^+)$ represents the generalized stresses on top surface for the interface H_i, and $\mathbf{V}_\mathbf{i}(\xi_y, H_i^-)$ represents the generalized stresses on bottom surface for the interface H_i. For the generalized displacements $\mathbf{U}_\mathbf{i}(\xi_y, H_i^+)$ and $\mathbf{U}_\mathbf{i}(\xi_y, H_i^-)$, they have a similar meaning.

In summary, based on the ordinary differential matrix equation established by Eq. (51), the solution of the corresponding function of layered transversely isotropic poroelastic media in the transformed domain under moving loads can be deduced by the extended PIM (Wang and Ai, 2018). Therefore, the dynamic response of the medium in physical domain can be further derived by using Eq. (23) and the inverse Fourier transform.

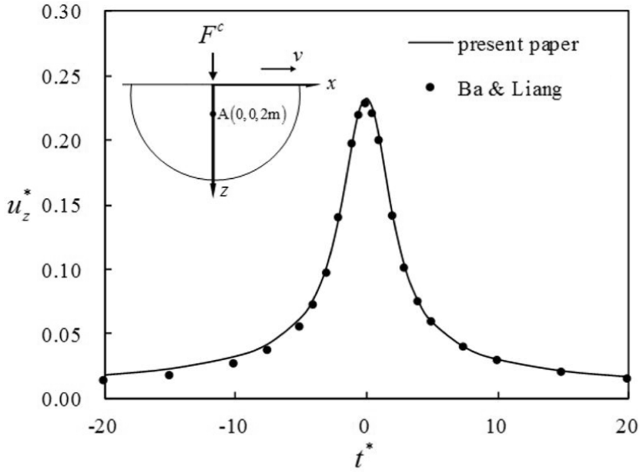

Fig. 3. Comparison of u_z^* at the calculation point$(0,0,2\text{m})$.

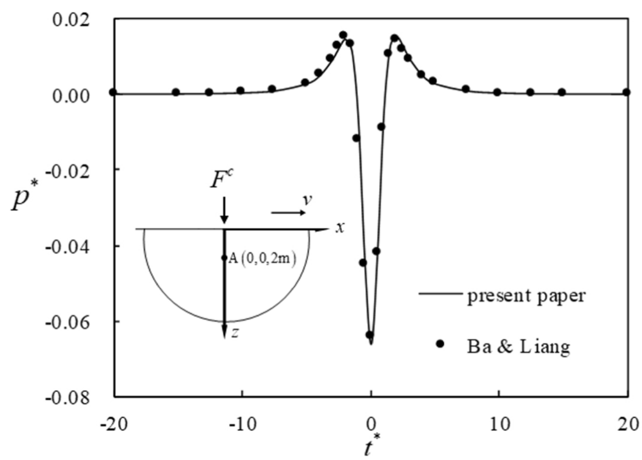

Fig. 4. Comparison of p^* at the calculation point$(0,0,2\text{m})$.

4. Verification

The solution in physical domain of the dynamic response is obtained through the integral transform inversion, and Gauss-Legendre numerical integration method is used in the programming for the realization of infinite integration in the inverse Fourier transform.

In order to verify the proposed solution, the results of Ba and Liang (2017) is compared in this section. Ba and Liang (2017) presented the vertical displacement u_z^* at point $\text{A}(0,\ 0,\ z)$ versus dimensionless time t^* inside the surface of an isotropic poroelastic half-space excited by a concentrated moving load with a constant speed. The dimensionless forms of time, displacement and velocity are $t^* = t\sqrt{G_0/\rho_0}/z$, $u_z^* = (G_0 z/F)u_z$ and $c^* = c/\sqrt{G_0/\rho_0}$, respectively, where $G_0 = 3.7$ GPa, $\rho_0 = 2155\ kg/\text{m}^3$ and $z = 2$ m. The relevant parameters are as follows: $c^* = 0.5$, $G_v = 3.7$ GPa, $E_h = 12.33$ GPa, $E_v = 6.165$ GPa, $\mu_h = \mu_{vh} = 0.25$, $\rho_s = 2650\ kg/\text{m}^3$, $\rho_f = 1000\ kg/\text{m}^3$, $\varphi = 0.3$, $K_s = 36.00$ GPa, $K_f = 2.00$ GPa, $\varsigma = 0.01$, $a_{\infty 1} = a_{\infty 3} = 2.167$. In Eqs. (12)–(14), the value of $\rho_f g/k_i\ (i = 1,3)$ is taken as $10^{-6}\text{Pa}{\cdot}\text{s/m}^2$. The calculation results of this paper are illustrated in Fig. 3, which are in good consistent with those of Ba and Liang (2017).

The pore pressure results calculated by the present solution are further compared with those of Ba and Liang (2017). The pore pressure p^* in $(0,0,z)$ is calculated here. The dimensionless form of pore pressure is defined as $p^* = pz^2/F$. As shown in Fig. 4, the results in this study are consistent with those of Ba and Liang (2017) which further proves the correctness of the proposed solution.

Table 1
Elastic modulus and shear modulus in different transversely isotropic parameters.

	$n = E_h/E_v$	E_h(Pa)	E_v(Pa)
Case1	0.5	1.0×10^8	2.0×10^8
Case2	1.0	2.0×10^8	2.0×10^8
Case 3	1.5	3.0×10^8	2.0×10^8
Case 4	2.0	4.0×10^8	2.0×10^8

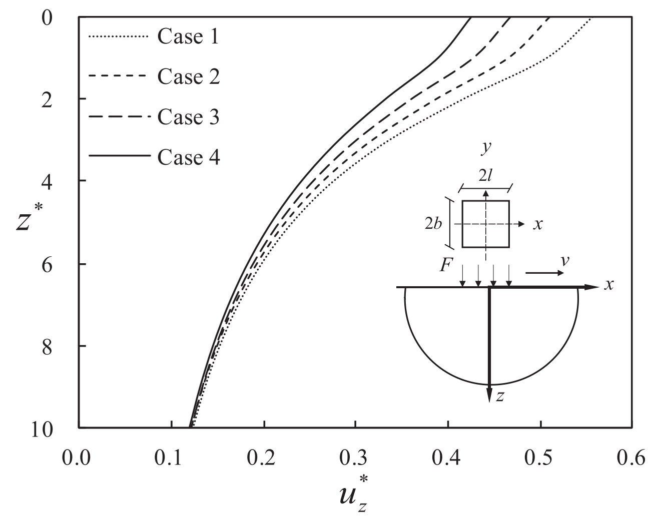

Fig. 5. Impact of the transverse isotropy on u_z^* along z-axis.

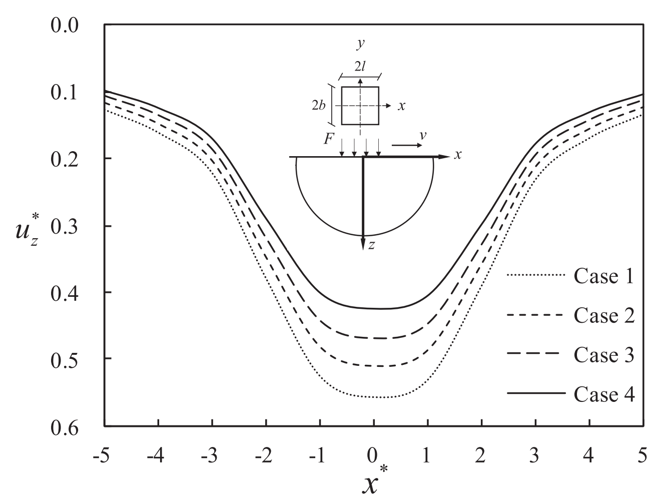

Fig. 6. Impact of the transverse isotropy on u_z^* along x-axis.

5. Numerical analyses

5.1. Effect of the transverse isotropy

An example is performed to analyse the impact of the transversely isotropic parameter $n = E_h/E_v$. A uniform rectangular moving load with intensity $F = 400kN/\text{m}^2$ and dimensions $2l = 2b = 4\text{m}$ is set to act on the surface of the medium. The dimensionless forms of vertical displacement and velocity are $u_z^* = u_z G_0/(Fl)$, $v^* = v/\sqrt{G_0/\rho_0}$, respectively, where $G_0 = 6 \times 10^7$ Pa and $\rho_0 = 1990kg/\text{m}^3$. The dimen-

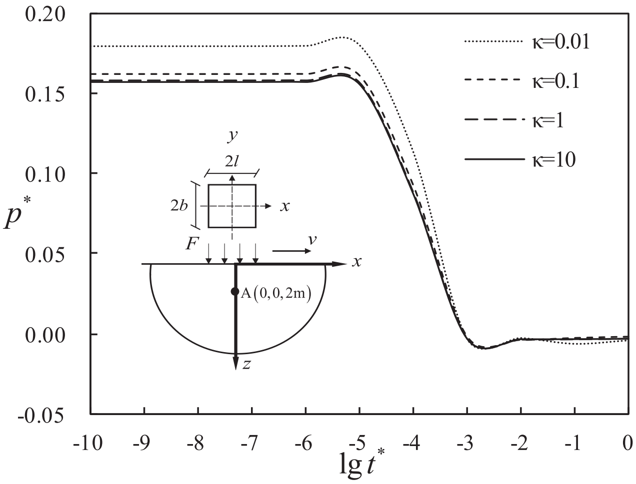

Fig. 7. Pore pressure p^* with different permeable anisotropy (point A).

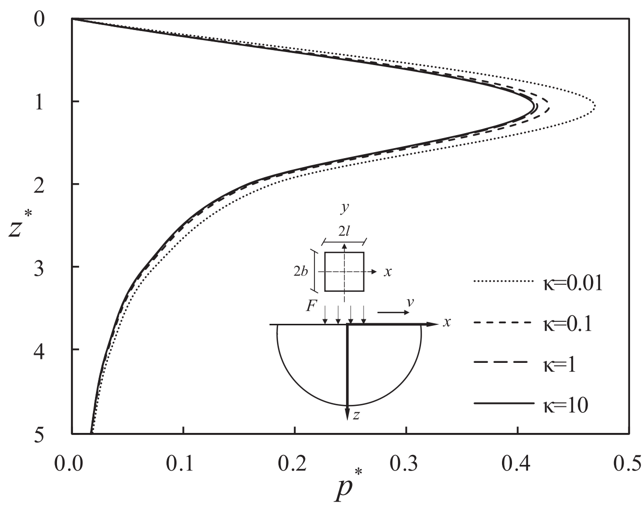

Fig. 8. Pore pressure p^* along z-axis with different permeable anisotropy.

sionless forms of length along z-axis and x-axis are $z^* = 2z/l$ and $x^* = 2x/l$. Let $v^* = 0.1$, $\mu_h = \mu_{vh} = 0.35$, $\rho_s = 2650kg/\text{m}^3$, $\rho_f = 1000kg/\text{m}^3$, $\varphi = 0.4$, $k_1 = k_3 = 10^{-7}m/\text{s}$, $K_s = 36.00$ GPa, $K_f = 2.00$ GPa, $\varsigma = 0.01$ and $a_{\infty 1} = a_{\infty 3} = 1.75$. Four cases are designed here to discuss the effect of the transversely isotropic parameter n. The values of n and the corresponding elastic modulus E_h, E_v in different cases are shown in Table 1. The displacement u_z^* of the medium along z-axis and x-axis of four cases are shown in Figs. 5 and 6, respectively.

It is obvious that u_z^* gradually decreases along the depth. As presented in Fig. 5, u_z^* decreases significantly with the increase of parameter n. The reason of the above phenomenon is that the rigidity of the medium increases due to the increase of E_h.

It can be observed from Fig. 6 that the maximum of u_z^* along the surface happens at the position where the moving load acts. Besides, u_z^* of the medium surface decreases overall as the parameter n increases.

5.2. Effect of the anisotropic permeability

An example is put forward in this section to illustrate the effect of anisotropic permeability $\kappa = k_1/k_3$. A single-layered transversely isotropic poroelastic medium with $\kappa = 0.01,\ 0.1,\ 1.0,\ 10.0$ is studied here. To simplify the analysis, the dimensionless time $t^* = tE_0k_0/\left(\gamma_w l^2\right)$

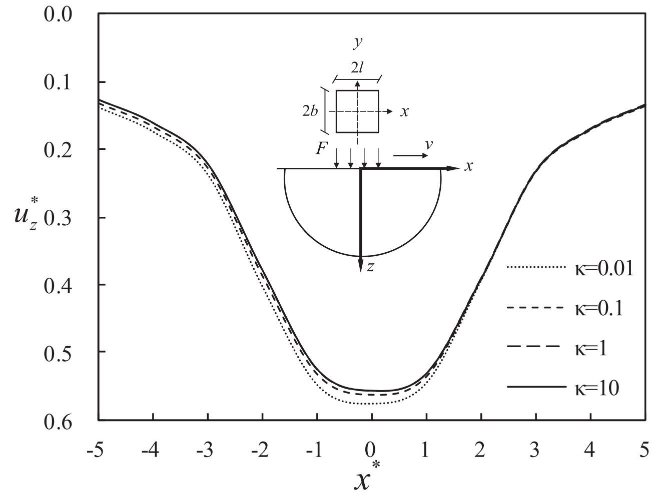

Fig. 9. Pore pressure p^* along x-axis with different permeable anisotropy.

Table 2
Parameters of different cases.

	Layer	$E_h(Pa)$	$E_v(Pa)$	h
Case 1	Layer 1	3.0×10^8	6.0×10^8	$3l$
	Layer 2	1.5×10^8	3.0×10^8	$3l$
	Layer 3	1.0×10^8	2.0×10^8	∞
Case 2	Layer1	6.0×10^8	12.0×10^8	$3l$
	Layer 2	1.5×10^8	3.0×10^8	$3l$
	Layer 3	1.0×10^8	2.0×10^8	∞
Case 3	Layer1	3.0×10^8	6.0×10^8	$3l$
	Layer 2	4.5×10^8	9.0×10^8	$3l$
	Layer 3	1.0×10^8	2.0×10^8	∞
Case 4	Layer1	6.0×10^8	12.0×10^8	$6l$
	Layer 2	1.5×10^8	3.0×10^8	$3l$
	Layer 3	1.0×10^8	2.0×10^8	∞
Case 5	Layer1	3.0×10^8	6.0×10^8	$3l$
	Layer 2	4.5×10^8	9.0×10^8	$6l$
	Layer 3	1.0×10^8	2.0×10^8	∞

and dimensionless pore pressure $p^* = p/F$ are defined, where $E_0 = E_v = 2 \times 10^8 Pa$, $k_0 = k_1 = 10^{-7}\text{m/s}$ and $G_v = 6.0 \times 10^7 Pa$. The parameters of moving load and the medium are the same with Case 1 in Section 5.2. Figs. 7–9 show the variation of pore pressure p^* with time at A(0, 0, 2m), the curves of pore pressure p^* along z-axis and the displacement u_z^* of the medium surface along x-axis, respectively.

It can be observed from Fig. 7 that the initial p^* at the same point decreases with the increase of $\kappa = k_1/k_3$, while p^* will dissipate quickly and eventually decrease to 0. The reduction of the vertical permeability coefficient k_3 affects the drainage consolidation process of the medium, but it has a negligible effect on the final consolidation state in the medium. In addition, the Mandel-Cryer effect appears in the process of consolidation, and this phenomenon becomes more obvious as the parameter κ decreases.

Because of the assumption that the top of the medium is permeable, p^* on that surface is 0. According to the curves from Fig. 8, p^* achieves the peak value in the shallow-layer medium. As κ increases, p^* at the same depth decreases, while it tends to be stable as the depth increases.

It is shown in Fig. 9 that as κ increases, u_z^* of the free surface decreases. To sum up, the permeability anisotropy parameter κ has a little

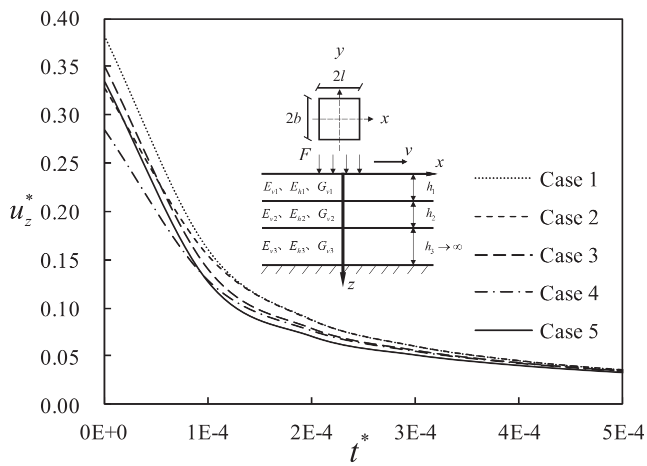

Fig. 10. Impact of the stratification on u_z^* with time.

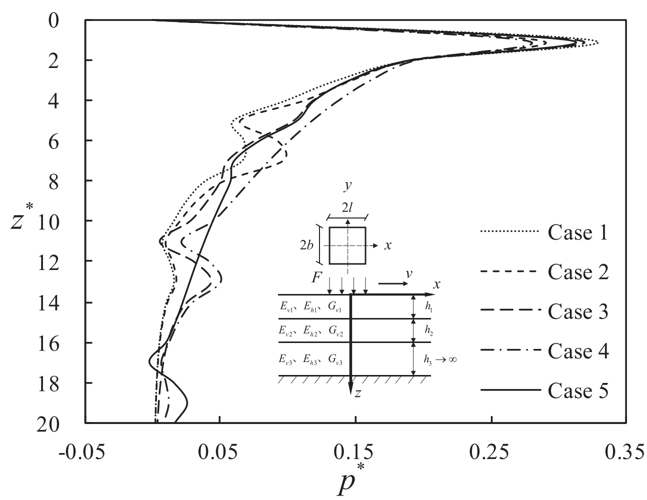

Fig. 11. Impact of the stratification on p^* along z-axis.

effect on p^* and u_z^* of the medium surface.

5.3. Effect of the stratification of the medium

In this section, a three-layered transversely isotropic poroelastic medium model is established. Five cases are designed to research the effect of stratification. The load condition is the same as that in Section 5.1. $G_v = 6.0\times 10^7 Pa$, $\mu_h = \mu_{vh} = 0.35$, $\rho_s = 2560 kg/\mathrm{m}^3$, $\rho_f = 1000 kg/\mathrm{m}^3$, $\varphi = 0.4$ and $k_1 = k_3 = 10^{-7}\mathrm{m/s}$. The other parameters of different cases are listed in Table 2. The results that show the change of the displacement u_z^* with time at coordinate origin and the change of the pore pressure p^* along z-axis are shown in Figs. 10 and 11, respectively.

It is observed from Fig. 10 that the elastic modulus and the thickness of the layer affects u_z^* of the medium surface a lot. The results of Case 1 and Case 2 show that as the elastic modulus of the medium increases, there is a certain decrease of u_z^* of the surface when $t = 0$. The same law can also be found from the results of Case 1 and Case 3. Moreover, by comparing Case 2 and Case 4, as well as Case 3 and Case 5, the conclusion can be obtained that the increase in the thickness of the medium layer with larger elastic modulus has a certain effect on reducing the surface displacement u_z^*.

Fig. 11 depicts the results of p^* along z-axis with different cases. We can find that the elastic modulus has a slight impact on p^*. However, the obvious fluctuations of curves shows that the stratification of the media has a great impact on p^*.

Table 3
Other parameters of the medium.

Layer	Thickness	E_h(Pa)	E_v(Pa)	G_v(Pa)	μ_h	μ_{vh}
Layer1	12 m	12.0×10^8	6.0×10^8	1.8×10^8	0.35	0.35
Layer 2	6 m	3.0×10^8	1.5×10^8	4.5×10^7	0.35	0.35
Layer 3	∞	6.0×10^8	3.0×10^8	9.0×10^7	0.35	0.35

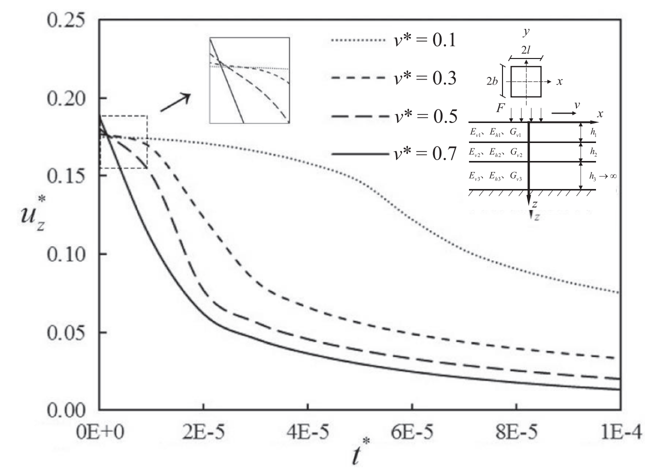

Fig. 12. u_z^* with different velocities.

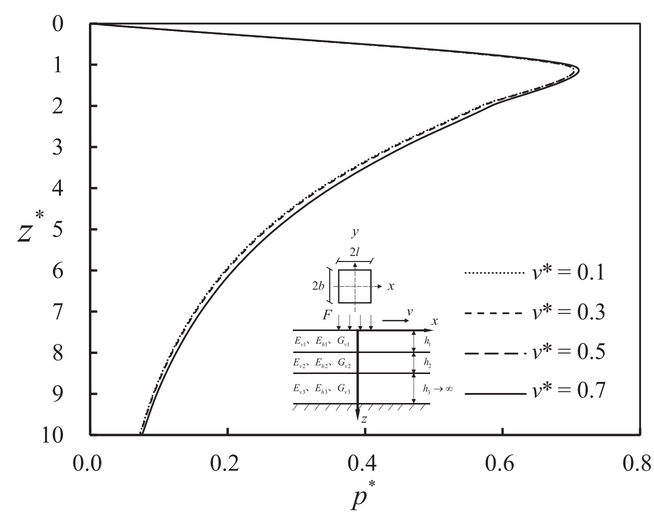

Fig. 13. p^* with different velocities along z-axis.

5.4. Effect of the speed of the moving load

To discuss the impact of the speed of moving loads, four cases are studied in this section. The form and dimensions of moving loads are the same with those in Section 5.1. This paper intends to simulate the working conditions of high-speed train or subway, so the speed range is set in the subsonic range, and the dimensionless velocity v^* of the moving load are 0.1, 0.3, 0.5 and 0.7. The parameters of the medium are selected as $\rho_s = 2560\mathrm{kg/m}^3$, $\rho_f = 1000\mathrm{kg/m}^3$, $\varphi = 0.4$ and $k_1 = k_3 = 10^{-7}\mathrm{m/s}$. Other parameters are listed in Table 3. Then the displacement u_z^* calculated by the present solution at coordinate origin with time, p^* along z-axis and u_z^* along x-axis are shown as in Figs. 12–14.

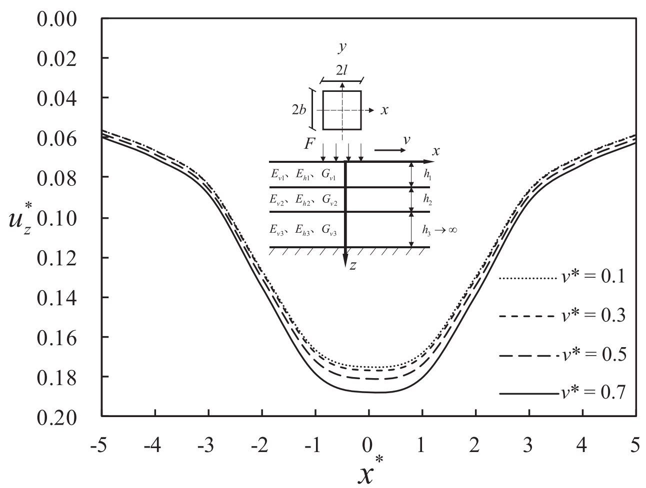

Fig. 14. u_z^* with different velocities along *x*-axis.

It can be noted from Fig. 12 that the higher the load speed, the larger the initial u_z^* at coordinate origin, and the steeper the curve. According to Fig. 13, it is shown that the increase of the load speed will lead to a small increase of p^* at any depth. Fig. 14 shows that the overall u_z^* of the medium surface will increase slightly with accelerating the moving load, which will not change the distribution characteristics of u_z^* of the medium surface.

6. Conclusions

By utilizing the extended PIM and Fourier transform technique, an exact solution of layered transversely isotropic poroelastic media under vertical rectangular moving loads is obtained. After verifying the accuracy of the proposed solution, several numerical examples are developed to discuss the influence of transverse isotropy, anisotropic permeability, medium stratification and moving load speed on the dynamic response of media. The results show that compared with the load velocity, the elastic modulus has a more significant impact on dynamic response of the medium. The Mandel-Cryer effect becomes more obvious as the permeable anisotropy becomes more remarkable. Besides, strengthening the surface layer of media is a relatively effective way to reduce the dynamic response of the media. This paper provides an effective method for estimating the displacement and excess pore fluid pressure of the media under rectangular moving loads, which would facilitate further research on the dynamic response excited by parabolic loads or more complex loads.

CRediT authorship contribution statement

Zhi Yong Ai: Conceptualization, Methodology, Writing - review & editing, Supervision. **Gan Lin Gu:** Data curation, Validation, Investigation, Writing - original draft. **Yong Zhi Zhao:** Investigation, Writing - review & editing. **Jun Jie Yang:** Data curation, Validation, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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